—floating, spawning, evolving? What strange beasts once swum there?

Whatever the dramas of those distant times, it all started to go wrong around 3.8 billion years ago. We can see that the erosion of ancient craters dramatically began to slow about then. As the atmosphere thinned, as the rivers flowed no more, as the oceans began to dry, as the temperatures plummeted, life would have retreated to the few remaining congenial habitats, perhaps huddling at the bottom of ice-covered lakes, until it too vanished and the dead bodies and fossil remains of exotic organisms—built, it might be, on principles very different from life on Earth—were deep-frozen, awaiting the explorers who might in some distant future arrive on Mars.

Meteorites are Fragments of other worlds recovered on Earth. Most originate in collisions among the numerous asteroids that orbit the Sun between the orbits of Mars and Jupiter. But a few are generated when a large meteorite impacts a planet or asteroid at high speed, gouges out a crater, and propels the excavated surface material into space. A very small fraction of the ejected rocks, millions of years later, may intercept another world.

In the wastelands of Antarctica, the ice is here and there dotted with meteorites, preserved by the low temperatures and until recently undisturbed by humans. A few of them, called SNC (pronounced “snick”) meteorites* have an aspect about them that at first seemed almost unbelievable: Deep inside their mineral and glassy structures, locked away from the contaminating influence of the Earth’s atmosphere, a little gas is trapped. When the gas is analyzed, it turns out to have exactly the same chemical composition and isotopic ratios as the air on Mars. We know about Martian air not just from spectroscopic inference but from direct measurement on the Martian surface by the *Viking* landers. To the surprise of nearly everyone, the SNC meteorites come from Mars.

Originally, they were rocks that had melted and refrozen. Radioactive dating of all the SNC meteorites shows their parent